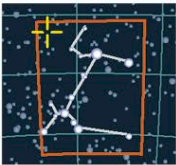


WOLF-RAYET STAR

WR 7



DISTANCE FROM SUN
15,000 light-years
MAGNITUDE 11.4
SPECTRAL TYPE WN5

CANIS MAJOR

The emission nebula NGC 2359, which has a diameter of around 30 light-years, has been produced by an extremely hot Wolf-Rayet star, visible at its center. This star, designated WR 7, has a surface temperature of between 54,000°F (30,000°C) and 90,000°F (50,000°C)—6 to 10 times as hot as the Sun. It is also highly unstable, ejecting stellar material into the interstellar medium at speeds approaching 4.5 million mph (7.2 million kph). Even though it is a massive star of around 10 solar masses, it is losing about the equivalent of the mass of the Sun every thousand years. With this level of mass loss, Wolf-Rayet stars like WR 7 are unable to exist in this stage of their life for long and are therefore rarely observed: only about 550 such stars are known in the Milky Way. Material



THOR'S HELMET

The popular name for the nebula surrounding WR 7 is Thor's Helmet, because it looks like a helmet with wings (above). The nebulae surrounding Wolf-Rayet stars are sometimes called bubble nebulae, and WR 7 lies at the center of the nebula's main bubble of hot gas. (The star is above and to the right of center in the image to the right.)

from the star has been ejected in an even, spherical manner, producing a bubble of material. This bubble has been further shaped by interactions with the surrounding interstellar medium. WR 7 is unusual because it lies at the edge of a dense, warm molecular cloud, and the asymmetrical shape of the outer parts of the surrounding nebula is due to "bow shocks," produced when fast stellar winds hit denser, static material.



STELLAR FIREBALL

WR 124 can be seen as a glowing body at the center of a huge, chaotic fireball. The fiery nebula surrounding the star consists of vast arcs of glowing gas violently expanding outward into space.

WOLF-RAYET STAR

WR 124



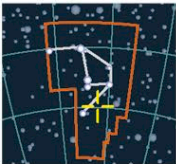
DISTANCE FROM SUN
15,000 light-years
MAGNITUDE 11.04
SPECTRAL TYPE WN

SAGITTARIUS

With a surface temperature of around 90,000°F (50,000°C), WR 124 is one of the hottest known Wolf-Rayet stars. This massive, unstable star is blowing itself apart—its material is traveling at up to 90,000 mph (150,000 kph). The observed state of M1-67, the relatively young nebula surrounding WR 124, is only 10,000 years old, and it contains clumps of material with masses about 30 times that of Earth and diameters of 90 billion miles (150 billion km).

PLANETARY NEBULA

Stingray Nebula



CATALOG NUMBER
Hen-1357
DISTANCE FROM SUN
18,000 light-years
MAGNITUDE 10.75

ARA

The Stingray Nebula is the youngest known planetary nebula. Observations made in the 1970s revealed that the dying star at the center of the nebula was not hot enough to cause the surrounding gases to glow. By the 1990s, further observations had shown that the central star had rapidly heated up as it entered the final stages of its life, causing the nebula to shine. This afforded astronomers a remarkable opportunity to observe the star in an exceedingly brief phase of its evolution. Because of its young age, the Stingray Nebula is one-tenth the size of most planetary nebulae, with a diameter only about 130 times that of the solar system. A ring of ionized oxygen surrounds the central star, and bubbles of gas billow out in opposite directions above and below the ring. Material traveling rapidly outward from the central star has opened holes

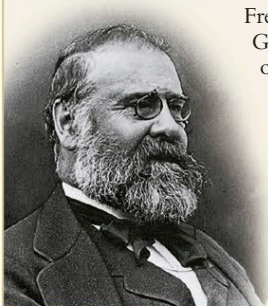
GRACEFUL SYMMETRY

The graceful, symmetrical shape of this very young planetary nebula gives it its popular name. In this enhanced true-color image, the Stingray Nebula's central star has a companion star just visible above it to the left.



in the ends of the bubbles, allowing streams of gas to escape in opposite directions. On the outer edges of the nebula, the central star's winds crash into the walls of the gas bubbles, generating shock waves and heat that cause the gas to glow brightly.

CHARLES WOLF AND GEORGES RAYET



French astronomers Charles Wolf (1827–1918) and Georges Rayet (1839–1906) co-discovered the type of unusual, hot stars that now bear their name. In 1867, they used the Paris Observatory's 16-in (40-cm) Foucault telescope to discover three stars whose spectra were dominated by broad emission lines rather than the usual narrow absorption lines (see pp.254–255). Today, over 500 Wolf-Rayet stars are known in our galaxy. Rayet later became Director of the Bordeaux Observatory.

GEORGES RAYET